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About the [UltraWave24 MWModulatedSource](#) : Features

Features

Basic functional specifications for [UltraWave24 MWModulatedSource](#) are:

- [Frequency range: 76MHz to 6GHz](#)

- Modulation bandwidth: 80MHz
- Amplitude range:
 - 76MHz to 2.7GHz: -130dBm to +13dBm
 - 2.7GHz to 6GHz: -130dBm to +10dBm
- Includes I and Q AWGs to support analog and digital modulation schemes
- Supports IQ signal definition in TSD files
- Supports IF signal definition for Gen4 MW compatibility
- Amplitude control of the modulated RF signal
- Signal power adjustment based on the crest factor of the modulation signal

For UltraWave24 hardware specifications, see [UltraWave24 Specifications](#).

UW24ModSrcAbout.1.3

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About the [UltraWave24 MWMModulatedSource](#) : Safety Information

Safety Information

When operating the Microwave Modulated Source instrument, always follow general UltraFLEX test system safety guidelines. See [Test System Safety](#).

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About the UltraWave24 MWMModulatedSource : Key Terms

Key Terms

AWG	Arbitrary Waveform Generator
baseband	Low frequency information contained in a modulated signal.
I signal	In-phase (real) component of a baseband signal.
IF	Intermediate frequency
LO	Local Oscillator
Q signal	Quadrature (imaginary) component of a baseband signal.
RF	Radio frequency

For more MW terms and definitions, see [Key Terms](#) in About the UltraWave24 Subsystem.

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About the UltraWave24 MWModulatedSource

About the UltraWave24 MWModulatedSource

The MWModulatedSource instrument generates modulated signals in the RF frequency band in the frequency range of 76MHz to 6GHz. Applications for the Microwave Source include prescalers, RF amplifiers, TV tuners, cellular radio, IF devices, and operational amplifiers. This section includes the following topics:

- | | |
|---|------------------------------------|
| • | Features |
| • | Safety Information |
| • | Key Terms |

Microwave Modulated Source information includes the following sections:

- | | |
|---|---|
| • | UltraWave24 MWModulatedSource Theory of Operation |
| • | UltraWave24 MWModulatedSource Programming |
| • | UltraWave24 MWModulatedSource Debug Display |

Related Topics

- | | |
|---|--|
| • | UltraWave24 Specifications |
|---|--|

- [MWMModulatedSource](#) (VBT syntax)
- [UltraWave24 DIB Design Guidelines](#)
- [About the UltraWave24 Subsystem](#)
- [UltraWave24 Examples](#)

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UltraWave24 MWModulatedSource Debug Display

UltraWave24 MWModulatedSource Debug Display

The Microwave Debug Display can display one or more instrument blocks for the MWModulatedSource logical instrument. Each instrument block shows the state of an instrument in the context of a site.

This following topic is available:

- MW24 MWModulatedSource Debug Display Controls

For an overview of all displays supporting the MW subsystem, see [UltraWave24 Displays](#).
For more information on using debug displays, see the following:

- Using Debug Displays in TDE
- Producing Code from Debug Displays

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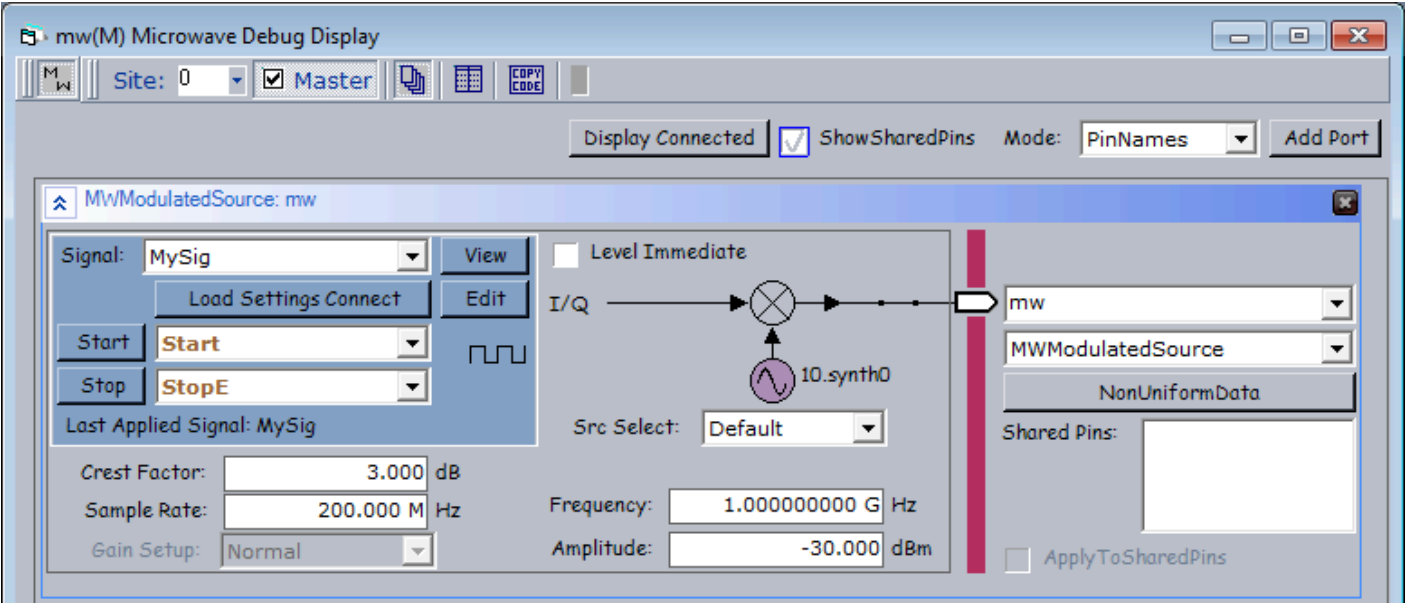


UltraWave24 MWModulatedSource Debug Display : MW24 MWModulatedSource Debug Display Controls

MW24 MWModulatedSource Debug Display Controls

MWModulatedSource port display controls allow you to view and set instrument parameters, which are also accessible through the Pins interface of the MWModulatedSource VBT language.

For information about the common debug display controls, see UltraWave24 Displays.



MW24 MWModulatedSource Logical Instrument Display

The following controls are available on the MWModulatedSource logical instrument block:

Frequency	Displays and sets the modulated signal frequency. The tooltip displays the allowable frequency range.
Amplitude	Displays and sets the modulated signal amplitude. The tooltip displays the allowable amplitude range.

Src Select	Displays and sets the synthesizer used to generate modulated source signals.
Level Immediate	Sets the MWModulatedSource leveling mode to Level Immediate, when checked. See MWModulatedSource.Pins.LevelImmediate in the VBT Syntax Reference.
Sample Rate	Displays and sets the modulator sample rate for the modulation signal. The tooltip displays the allowable sample rate range.
Crest Factor	Displays and sets the crest factor for a modulated signal. The tooltip displays the allowable crest factor range.
Gain Setup	Displays and sets the gain setup for a modulated tone. For the UltraWave24 MWModulatedSource, This control is grayed out.
synth label	Identifies the synthesizer (slot and number) used by this MWModulatedSource logical instrument.
Signal	Name of the signal used by the source. Available signals appear on the drop-down list. If a default signal has been defined, the default signal's name appears automatically in this field. If the menu contains no signal names, no signals were previously defined in VBT.
View	Displays a list of signal viewer tools (such as WaveScope or VectorScope) for the selected Signal on a drop-down list of available viewers.
Load Settings Connect	Loads the selected signal's timing settings as specified on the Mixed Signal Timing sheet, as well as all of the instrument's settings and connects the MWModulatedSource.
Edit	Brings up the PSet and Signals Editor (PSSE) with the selected signal loaded. This control is grayed out if no signals are available. See PSet and Signals Editor in the TDE section.
Info	Displays property values for the selected Signal in a pop-up balloon window.
Start	Starts the selected Signal using one of the start options: - Start - Start1

- [StartE](#)

- [Start1E](#)

Stop

Stops the selected Signal using one of the stop options:

- [Stop](#)

- [StopE](#)



The square wave is a sourcing indicator that moves when the source is sourcing on the tester. This feature only works on line.

Last Applied Signal

Displays the name of the last signal loaded to the MWModulatedSource.

connection relay

The relay control connects or disconnects the MWModulatedSource logical instrument to the specified pin or channel. Click the relay to open or close the connection.



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UltraWave24 MWModulatedSource Programming

UltraWave24 MWModulatedSource Programming

The MWModulatedSource is a MW logical instrument in the IG-XL environment. You add the MWModulatedSource to an IG-XL program in the same general way that you add other MW instruments.

The following topics are available:

- | | |
|--|--|
| | <ul style="list-style-type: none">Sourcing Modulated Signals Using the MWModulatedSource |
| | <ul style="list-style-type: none">MWModulatedSource Pattern Control |

Before writing test code using the MWModulatedSource, see [UltraWave24 Instrument Programming](#).
For more information, see:

- | | |
|--|--|
| | <ul style="list-style-type: none">MWModulatedSource (VBT Syntax Reference) |
| | <ul style="list-style-type: none">UltraWave24 Examples |

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UltraWave24 [MWModulatedSource Programming](#) : Sourcing Modulated Signals Using the MWModulatedSource

Sourcing Modulated Signals Using the MWModulatedSource
The following topics are available:

- | | |
|---|--------------------------------------|
| • | Programming Concepts |
| • | Programming Steps |

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UltraWave24 MWModulatedSource Programming : Sourcing Modulated Signals Using the MWModulatedSource : Programming Concepts

Programming Concepts

TSD File Support

The MWModulatedSource accepts signal definitions for modulated signals in TSD files. TSD files are loaded using the .SignalDefinitionFile property for the instrument.

For more information, see:

- Using the ESA Toolkit
- TheHdw.MWModulatedSource.Pins.Signals.Item.SignalDefinitionFile

MWModulatedSource Reset Values

The table lists local reset values for the MWModulatedSource.

MWModulatedSource Reset Values

Setting	Reset Value
Amplitude	-30dBm
Frequency	1.0GHz
Sample Rate	133.333MHz
Crest Factor	3.00dB
Connections	Disconnected

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UltraWave24 MWModulatedSource Programming : Sourcing Modulated Signals Using the MWModulatedSource : Programming Steps

Programming Steps

To source a modulated signal to an assigned RF port:

1. Create a modulation signal in VBT or import a signal waveform from an external file.

Use the Signals interface for the AWG instrument to specify all parameters of your modulation signal. Use the Wave Definitions sheet to specify your wave shape.

2. Specify the crest factor. Set up the MWModulatedSource with:

– Amplitude

– Frequency

– IF Frequency

– Crest factor

3. Connect the MWModulatedSource.

For an example of MWModulatedSource programming, see MWModulatedSource-MWReceiver AM Signal Test.

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UltraWave24 MWModulatedSource Programming : MWModulatedSource Pattern Control

MWModulatedSource Pattern Control

You can control an MWModulatedSource instrument using pattern microcodes, allowing changes to instrument states and parameters during a program. You can obtain repeatable timing by programming an instrument using pattern microcodes and Signals programming. You create a pattern independently of the test program using the PatternTool or a text editor. Patterns are composed of a series of numbered digital vectors. These digital vectors define the data for each channel listed in the pattern, as well the pattern microcode used for controlling the instrument during pattern execution. The following topics are available:

- MWModulatedSource Pattern Microcodes
- MWModulatedSource ASCII Pattern Syntax

Note:

To use pattern microcodes to control an MWModulatedSource on UltraFLEX, you must include a digital pin in the pin map, the channel map, and the pin list of the pattern module.

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UltraWave24 MWMModulatedSource Programming : MWMModulatedSource Pattern Control :
MWMModulatedSource Pattern Microcodes

MWMModulatedSource Pattern Microcodes

A microcode controls a single function, such a starting or stopping the sourcing of a signal. Extended microcode commands are available for use inside a pattern. Patterns allow you to control the precise timing of when a signal is sourced. To establish repeatable timing, or to synchronize with other instruments (such as the digital subsystem) in the tester, use a pattern. If you want to start generating a signal but are not concerned with precise timing, however, then you do not need a pattern.

The table lists the extended microcodes used to issue commands to an MWMModulatedSource instrument from a pattern.

MWMModulatedSource Pattern Microcodes

Microcode	Functional Description
Nop	No operation.
Resync	Forces phase alignment of an MWMModulatedSource instrument clock with the HSD T0 clock. See Use of Resync.

Start <SignalName>	The instrument immediately (on the next available logical analog clock) starts sourcing the named signal in SMEM and repeats it continuously until a Stop microcode is executed, the instrument is reset, or another signal is started. This microcode takes an optional argument, which is the name of the signal to be sourced. If the signal name is not specified, the default signal is used (see TheHdw.MWModulatedSource.Pins.Signals.DefaultSignal).
StartE <SignalName>	At the end of the currently sourcing signal's segment, the instrument sources the named signal in SMEM and repeats it continuously until a STOP microcode is executed, the instrument is reset, or another signal is started. This microcode takes an optional argument, which is the name of the signal to be sourced. If the signal name is not specified, the default signal is used (see TheHdw.MWModulatedSource.Pins.Signals.DefaultSignal).
Start1 <SignalName>	The instrument immediately starts sourcing (on the next available logical analog clock) the named signal in SMEM and sources it exactly once. This microcode takes an optional argument, which is the name of the signal to be sourced. If the signal name is not specified, the default signal is used (see TheHdw.MWModulatedSource.Pins.Signals.DefaultSignal).
Start1E <SignalName>	At the end of the currently sourcing signal's segment, the instrument sources the named signal in SMEM and sources it exactly once. This microcode takes an optional argument, which is the name of the signal to be sourced. If the signal name is not specified, the default signal is used (see TheHdw.MWModulatedSource.Pins.Signals.DefaultSignal).
Stop	The instrument stops sourcing the currently sourced signal immediately. STOP does not impact a signal that is running once.
StopE	The instrument stops sourcing the currently sourced signal at the end of the signal.
LoadSettingsConnect <SignalName>	Instructs an MWModulatedSource to load the settings for the named capture signal and connects the hardware to the DUT.

	SignalName is required.
Disconnect	Instructs an MWModulatedSource to reset all connections to their disconnected state.
Enable_Inst_Cond	Enables the instrument condition. The MWModulatedSource asserts the instrument condition to the pattern generator at the completion of a capture.
Disable_Inst_Cond	Disables an MWModulatedSource from asserting the instrument condition.

Notes on MWModulatedSource Microcodes

Note the following when using microcodes with an MWModulatedSource:

- IG-XL has rules regarding the necessary spacing in vectors between microcodes on a specific channel. If you place microcodes too close together, IG-XL will return an error. General guidelines for spacing microcodes include the following:
 - Resync: <1s
 - LoadSettingsConnect: 400s (typical)
 - Disconnect: <200s (typical)
 - Start: <10s
- An MWModulatedSource does not use PSets. It provides the equivalent capability using Signals programming in VBT.
- When using an UltraWave24 logical instrument (such as an MWModulatedSource), any vector that contains a POP microcode must be fully defined across all defined logical instrument instances.
 - IG-XL returns an error if you do not follow this guideline.

- Reapplying the same Signal to a given logical instrument instance results in the hardware NOPing the application. (This prevents glitches.)

For more information and an example, see [Using UltraWave24 Instruments with POP](#).

Use of Resync

The Resync microcode in the UltraWave24 subsystem performs additional tasks when compared to the Resync used with the BBAC and VHFAC instruments. The primary function of Resync for BBAC and VHFAC is to align the analog clock for the analog instrument with the digital clock supplying the timing reference for the pattern.

For the Microwave instruments, the Resync microcode also:

- Forces alignment of the internal AWG clocks which drive I and Q modulation AWGs
- Synchronizes the start of waveforms

The instantiation time associated with Resync is 1 s, which you must manage. Therefore, you must place a wait of at least 1 s between the Resync and following microcode.

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UltraWave24 MWMModulatedSource Programming : MWMModulatedSource Pattern Control :
MWMModulatedSource ASCII Pattern Syntax

MWMModulatedSource ASCII Pattern Syntax

MWMModulatedSource pattern microcodes are instructions that you can place on an individual vector of a pattern. For that vector, a microcode instructs the MWMModulatedSource to perform a particular action, such as starting a new signal.

You can only use MWMModulatedSource microcodes with pins that the instruments statement has specified as MWMModulatedSource. For example:

```
instruments = {
    DUT_AIN:MWMModulatedSource;
}
```

For more information, see [Instruments Statement](#) in the Pattern Language section.

If a vector contains any MWMModulatedSource related microcode other than nop, that vector is placed in SRM.

The format for specifying an MWMModulatedSource microcode on a particular vector is:

(pin-list : MWMModulatedSource = microcode)

The following code shows an example of ASCII pattern microcode syntax for an MWMModulatedSource. The figure shows this pattern in PatternTool.

```
instruments = {
    Source:MWMModulatedSource 1;
}

vm_vector
vm_MWMModSourceOnly_A ( $tset , Dig1)
{
    start_label MWMModSourceOnly:
```

[illegible]

```
halt
}
```

MWMModSourceOnly.PAT (vm_MWMModSourceOnly_A - VM)						
TSet	Command	Label	Vector	Source (MWMModulatedSource)	D i g i t	Comment
BasicSource		start_label MWMModSourceOnly:	0		0	
BasicSource			1	Resync	0	
BasicSource	repeat 10		2		0	1uS wait time
BasicSource			3	LoadSettingsConnect GSM	0	
BasicSource	repeat 10000		4		0	1mS wait time
BasicSource			5	Start GSM	0	
BasicSource	repeat 100		6		0	10uS wait time
BasicSource			7	LoadSettingsConnect WCDMA	0	
BasicSource	repeat 10000		8		0	1mS wait time
BasicSource			9	Start WCDMA	0	
BasicSource	repeat 100		10		0	10uS wait time
BasicSource			11	Stop	0	
BasicSource	repeat 100		12		0	10uS wait time
BasicSource			13	Disconnect	0	
BasicSource	repeat 2000		14		0	200uS wait time
BasicSource			15		0	

Vertical: 65 Vectors | Vectors 0 to 15

MWMModulatedSource Pattern Microcodes Example

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UltraWave24 MWModulatedSource Theory of Operation

UltraWave24 MWModulatedSource Theory of Operation

The MWModulatedSource instrument produces a programmable modulated RF signal by modulating the output of an RF frequency synthesizer. The MWModulatedSource has built-in I and Q AWGs to support both analog and digital modulation schemes.

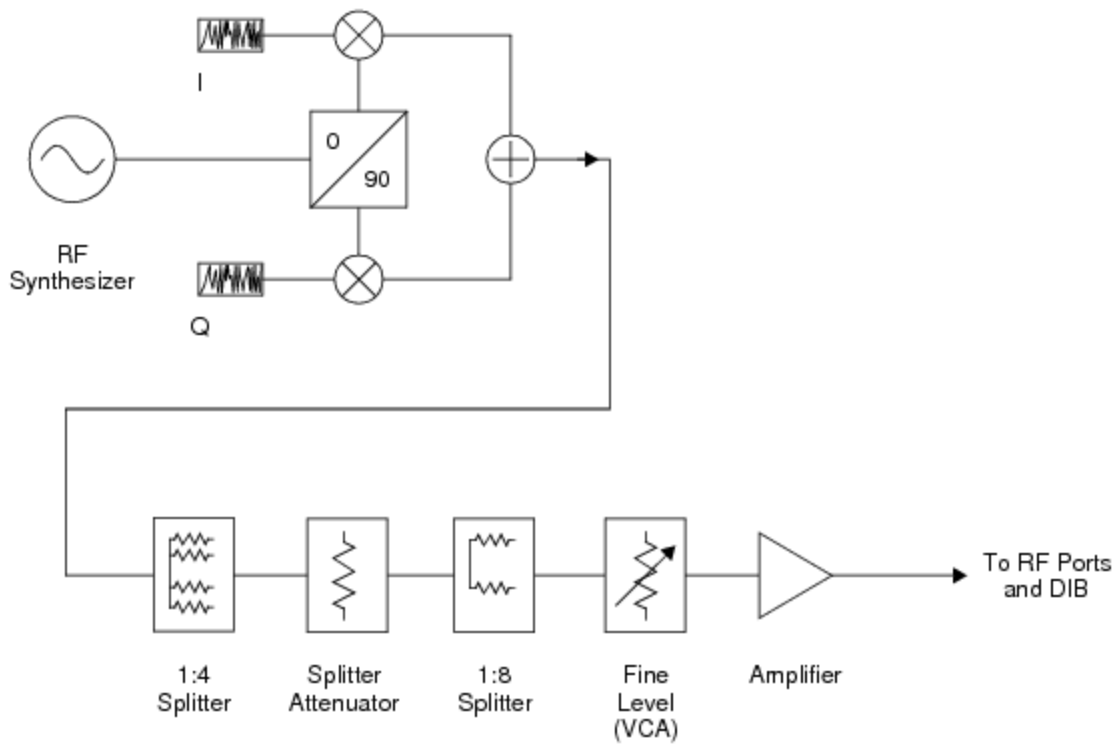
This following topics are available:

- | | |
|---|--|
| • | Basic Operation |
| • | IQ Signal Input and Conversion |
| • | MWModulatedSource Hardware Functions and Signal Flow |

For information on MWModulatedSource instrument performance specifications and standards, see [UltraWave24 Specifications](#).

Basic Operation

The following figure shows a basic functional model of the MWModulatedSource.



UltraWave24 MWModulatedSource Block Diagram (old)

As shown, the modulated RF signal is produced by mixing an LO signal from an RF frequency synthesizer with I and Q signals.

Each modulated source is split for up to four MWMeasureHD boards. Like other UltraWave24 sources, the MWModulatedSource passes through the splitter attenuator before being split for multiple front ends. Each source path has fine level control and amplifications.

Note:

The signal path of a MWModulatedSource has more than one separate attenuator and amplifier. All level control elements are combined in the blocks in the diagram.

The splitters in the signal path are part of the fanout architecture of the UltraWave24. They allow multiple MWModulatedSources to connect to multiple RF ports and device pins, without splitters on the DIB.

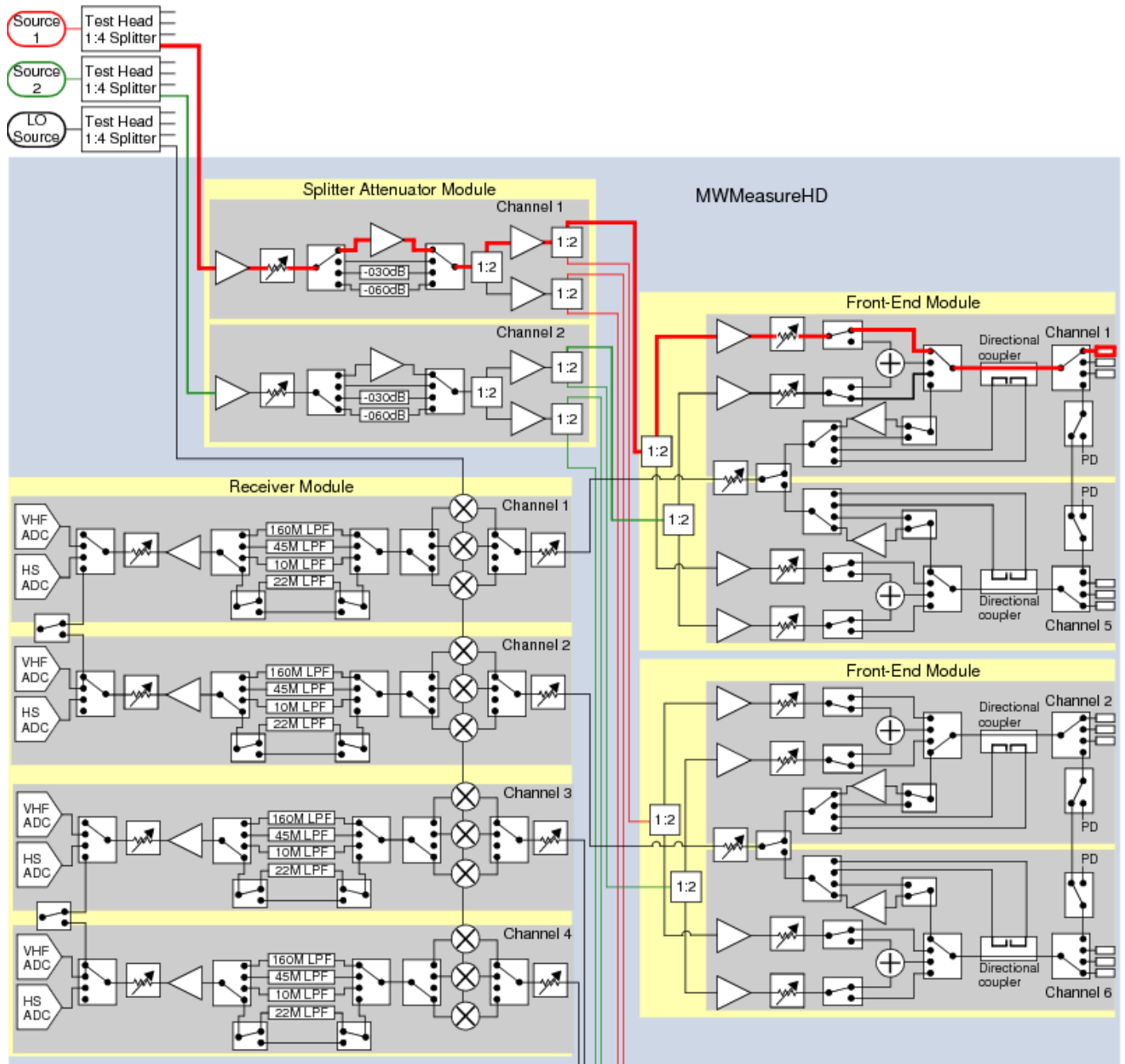
IQ Signal Input and Conversion

The MWModulatedSource uses input I and Q signals from the Synthesizer. I and Q signal data is loaded using the Signals interface.

Baseband signals specified as IF data are converted to IQ data by the instrument driver.

MWModulatedSource Hardware Functions and Signal Flow

The following figure shows the configuration of the UltraWave24 subsystem and signal flow for the operation and connection of a MWModulatedSource. Heavy lines represent the signal flow. See [UltraWave24 Hardware Functions](#) for more information about the functions in the diagram.



MWModulatedSource Hardware Functions and Signal Path, Two of Four FEMs Shown For DIB connections (MW ports, Pogo pins) and signal names for the MW subsystem and instruments, see [User View of UltraWave24 SMPM Connector Blocks](#), [User View of UltraWave24 DIB Pad Patterns](#), and [UltraWave24 Signal Descriptions](#).



